

Econophysics in Financial Markets

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Note: Originally written in October 2025 for the PC5253 Complex Systems Analysis and Modelling module at National University of Singapore (NUS). Updated in September 2026.

1. Introduction

Financial markets are complex networks of interactions between brokers, companies, and individuals. The markets facilitate the transactions or trades these parties make with each other, to exchange assets (such as commodities, currencies, bonds, and futures) for money. In 2024, the total value of the global financial market was USD 126.7 trillion, with equity issuance amounting to USD 504.8 billion [1]. The immense volume of capital traded on financial markets underscores the need for reliable methods to quantify and manage financial risk, thus the search for accurate models for predicting market behavior is an ongoing one. Econophysics, the application of statistical physics methods to study problems in economics [2], has emerged as a promising approach to model the dynamics of financial markets.

One of the key contributions of statistical physics to economics is the concept of power-law. Empirical observations of power-law behaviour in the changes of financial indices were reported in a 1999 study by P. Gopikrishnan and his team of researchers, which results in a significantly higher probability of extreme events compared to a Gaussian distribution [3]. In this report, we validated Gopikrishnan et al's findings in the current financial market by analyzing the data of two financial indices and one cryptocurrency and described the implications of our results.

2. Hypotheses

Gopikrishnan et al probed the power-law behaviour on the price changes of financial indices by examining the time series of returns of the financial indices across different time scales Δt . The return of a given asset is defined as the change in the natural logarithm of the asset price over a time interval Δt ,

$$r_t \equiv \ln(p_{t+\Delta t}) - \ln(p_t)$$

where p_t is the price of the asset. The first model of the stochastic process of returns was presented by Louis Bachelier in 1900—a random walk of independent, identically distributed (i.i.d.) random variables, therefore resulting in a Gaussian distribution of returns due to Central Limit Theorem [4]. While Bachelier's work has pioneered much research in the field of Econophysics [5], there were a number of limitations in his proposal. For example, his model predicts that extreme price changes were nearly impossible. On the contrary, empirical studies

[6] have found the presence of tails in the distribution of returns distinct from a Gaussian distribution. The data showed a high frequency of large events, a feature which is substantially underestimated in a Gaussian process.

Another pioneering study on returns, made by Benoit Mandelbrot in 1963 on cotton prices, noted that besides their non-Gaussian behaviour, the returns exhibit time-scaling, where the functional form of their distribution is similar at different time intervals Δt ranging from 1 day to 1 month [6]. These findings led him to propose that the distribution of returns can be modelled as a Levy stable distribution. Levy stable distributions originate from a generalized form of the Central Limit Theorem. They are distributions of i.i.d. random variables with infinite first or second moment, where the sum of these distributions retains the same functional form of the constituent distributions as the number of variables increases.

In 1999, motivated by a larger availability of financial datasets and the advances in computation, Gopikrishnan et al tested Mandelbrot's hypotheses empirically on datasets of three financial indices that indicate the market performance of various parts of the world (United States, Japan, and Hong Kong), for various Δt between 1 minute and 1 month [3]. They concluded that the distributions of returns exhibit power-law behavior at the tails with a characteristic exponent $\alpha \approx 3$ from $\Delta t = 1$ minute up to 1 month. The power-law distribution is defined as

$$P(|r_t| > x) \sim x^{-\alpha}$$

where $P(|r_t| > x)$ is the probability that a given return has an absolute value larger than x . The exponents found by Gopikrishnan et al being similar across time scales is consistent with the time-scaling reported in Mandelbrot (1963) but seems at odds with the supposed heterogeneity of the functional form of unstable distributions as sample size changes that Mandelbrot postulated. The reason for time-scaling, argued and then empirically proven in the 1999 paper, resulted from the slow decay of volatility correlations underlying the distribution of returns. Volatility being a measure of price fluctuations over time, the correlations described here refer to how prices continue to fluctuate to a similar degree over time.

The empirical studies in Gopikrishnan et al (1999) showed that the returns distribution cannot be modelled completely on the Levy stable distribution, but rather is consistent with a truncated Levy distribution. This model was proposed in 1995 by R. Mantegna and H. Stanley when they

noted that the central part of the distribution of S&P 500 returns is well-aligned to a Levy stable distribution but the asymptotic behaviour decays at a faster rate than predicted by Levy stable distributions. The truncated Levy distribution possesses a finite variance due to the roughly exponential truncation, therefore by traditional CLT this model converges eventually to Gaussian behaviour, with the Levy-like central region slowing down this convergence. The 1999 paper compared the moments of return distributions at various time scales and found a gradual convergence as predicted by the truncated Levy distribution.

A quarter of a century later, the advent of the information age and the widespread use of automatic trading algorithms have led markets to become more volatile. In such an economy, the analysis of up-to-date data by traders is essential to gain a competitive advantage over other market participants. In the following sections, we will therefore verify Gopikrishnan et al's findings empirically on more recent datasets to investigate their validity in the current financial market.

A practical application of modeling the distribution of returns lies in its potential to detect financial bubbles and anticipate market crashes. As discussed earlier in this section, Montegna & Stanley (1995) and Gopikrishnan et al (1999) have demonstrated that the return distribution can be represented by a Levy stable distribution for central values. A Levy stable distribution S of random i.i.d. variables x_i is provided by four parameters,

$$x \sim S(\alpha_S, \beta_S, \gamma_S, \delta_S)$$

where $\alpha \in (0, 2]$ is the stability parameter and $\beta \in (-1, 1)$ is the skewness parameter, $\gamma \in [0, \infty)$ is the scale parameter and $\delta \in (-\infty, \infty)$ is the location parameter. The distribution is symmetric when $\beta = 0$, right-skewed when $\beta = +1$, and left-skewed when $\beta = -1$. Bielinskyi et al (2019) showed that the α and β parameters from fitting the distribution of returns to a Levy stable distribution can describe the level of development of a stock market. Less developed stock markets tended to have reduced α and absolute value of β , which corresponded to a higher frequency and persistence of crises [9]. The parameters had also consistently decreased as the Dow Jones Industrial Average (DJIA) index entered crisis states between 1920 and 2019, encouraging the proposal by the authors that these parameters can serve as precursor-indicators of impending financial crises. Our analysis will test whether the parameters can signal an approaching crisis in other stock indices and in cryptocurrency.

The hypotheses guiding our empirical studies are to validate a) the power-law behavior of the asymptotic distribution of returns; b) that the returns starting from time scales $\Delta t = 1$ min onwards are not Levy stable distributions; c) the gradual convergence of the returns distribution towards a Gaussian distribution with a time scale equal to or larger than $\Delta t \approx 4$ days; and d) whether the stability α and asymmetry β move towards zero in the lead-up to a crisis.

3. Data Sources

We analyze the data of S&P 500 to verify the above hypotheses on recent financial markets. S&P 500 consists of the top 500 companies in the United States in terms of market value, liquidity, and industry representation. It serves as one of the most widely used measures of overall U.S. stock market performance. Closing price data recorded at 1-minute intervals between the years 2016 to 2024 was drawn from Histdata public data repository [\[10\]](#). To compare the returns distribution across time scales, data recorded on daily, several days, and monthly frequencies for the same span of time was drawn from NASDAQ public data repository [\[11\]](#).

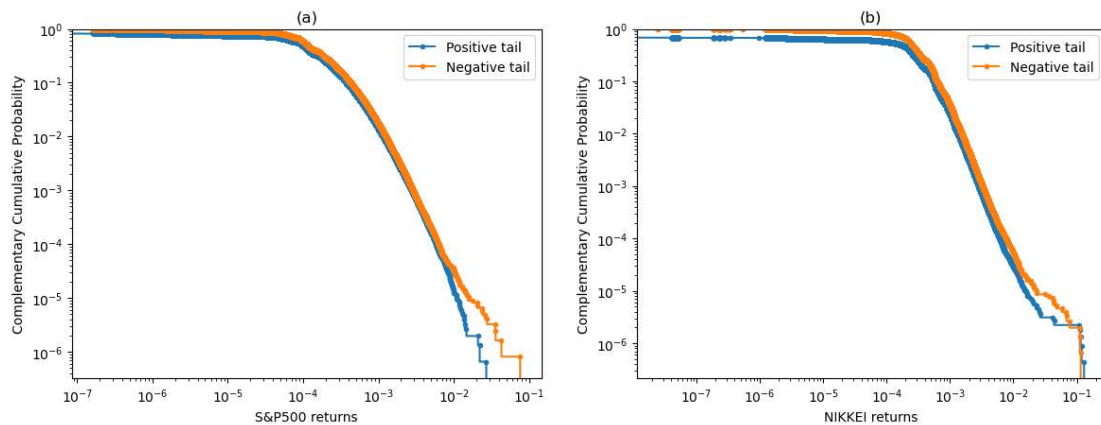
For an assessment of the robustness of our S&P 500 results, we did a comparative analysis on another financial index, NIKKEI, and a cryptocurrency asset, Bitcoin (BTC). NIKKEI tracks the performance of 225 Japanese companies on metrics like the S&P index. NIKKEI prices recorded at 1-minute intervals between the years 2010 to 2024 was drawn from the Histdata public data repository [\[12\]](#). Data recorded on daily, several days, and monthly frequencies for the same span of time was drawn from the NIKKEI public archives. As Gopikrishnan et al's study was focused on financial indices, Bitcoin— having the largest market size among cryptocurrency assets— was chosen to demonstrate whether the power-law distribution can be extrapolated to the returns of individual assets. The price data was sourced from the Binance public data repository (Binance Vision) with a temporal resolution of 1 millisecond and covering the period from October 1, 2024 to September 30, 2025 [\[13\]](#). Since Bitcoin did not experience crises in that time, we obtained Bitcoin price data recorded at 1-day frequency from a longer time period of 1 January, 2022 to 30 June, 2026 for our estimation of Levy stable distribution parameters on the Bitcoin returns distribution.

4. Methods of Analysis

To probe whether the asymptotic distributions of returns follow power-law, we made log-log plots of the complementary cumulative distribution of the returns for time intervals $\Delta t = 1$ min, 1 day, 4 days, and 10 days. Due to the finer time-resolution, the time intervals selected for the Bitcoin data were smaller, $\Delta t = 1$ s, 1 min, 1 day, and 4 days. The characteristic exponent α of the power-law tails were computed using power-law maximum likelihood estimate fitting (from the Python powerlaw package) for a comparison against the power-law fitted values obtained by Gopikrishnan et al and discerning if each returns distribution is in the Levy stable regime. The critical time scale at which the returns distribution starts converging towards a Gaussian one was determined by calculating the third (skewness) and fourth moments (Pearson's kurtosis) of the data and comparing the moments against those of a normal distribution. Estimation of the scaling and asymmetry parameters of the distributions of returns was performed by fitting 500 consecutive days of daily-frequency data (excluding off-trading hours) to a Levy stable distribution (using the SciPy Levy_stable function) on a grid of various days.

5. Results and Implications

The complementary cumulative probability distributions of the S&P 500, NIKKEI, and Bitcoin returns where $\Delta t = 1$ min are plotted in Figure 1. Each of the plots exhibit a linear relationship between the complementary cumulative probability $P(|r_t| > x)$ and the returns r_t when $r_t > 10^{-3}$, indicating that the asymptotic distribution of returns follows a power-law, which is in concurrence with the empirical findings of Gopikrishnan et al (1999) [3]. Similar behavior was observed when $\Delta t = 1$ day and 4 days, as plotted in Figure 2. The exponents of the power-law tails were calculated using power-law fitting and reported in Table 1.



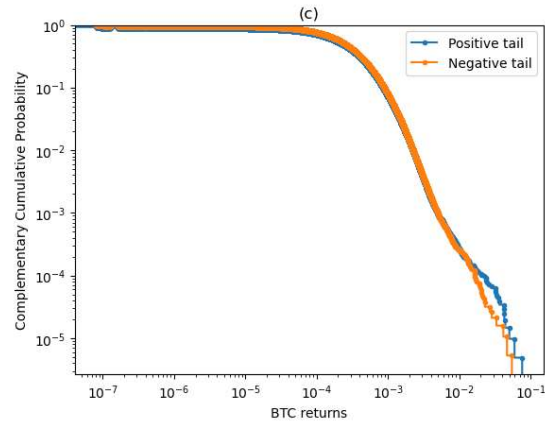
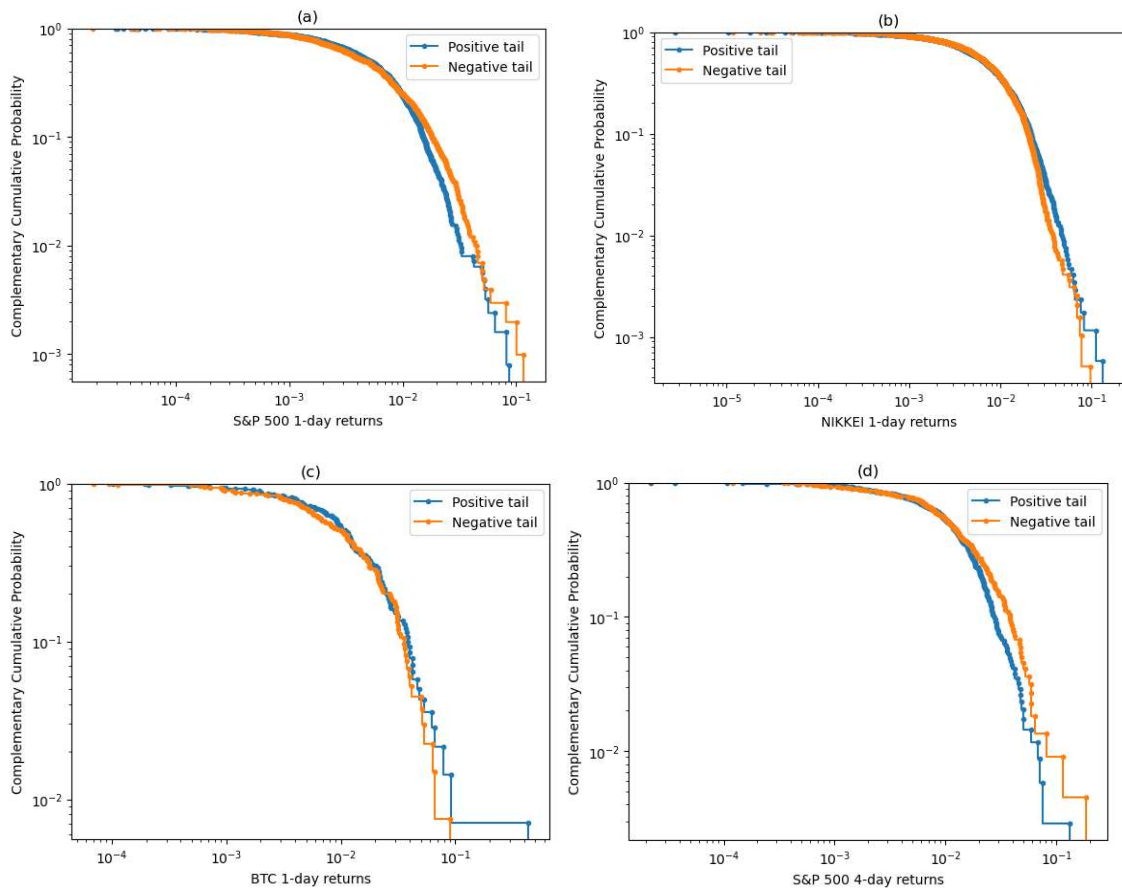


FIGURE 1. Log-log plots of the complementary cumulative probability distributions of (a) S&P500, (b) NIKKEI, and (c) Bitcoin returns when $\Delta t = 1$ min. Positive and negative returns are represented by the blue and orange lines respectively.



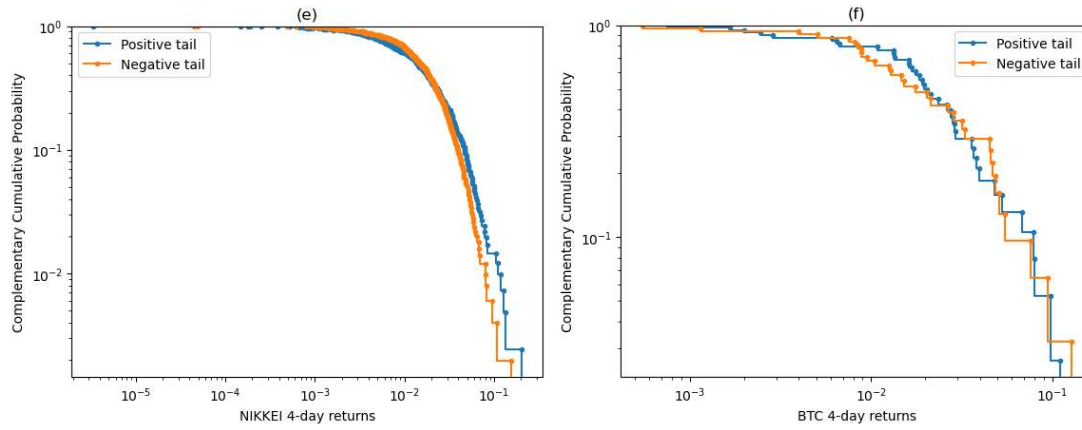


FIGURE 2. Log-log plots of the complementary cumulative probability distributions of (a) S&P500, (b) NIKKEI, and (c) Bitcoin returns when $\Delta t = 1$ day and (d) S&P500, (e) NIKKEI, and (f) Bitcoin returns when $\Delta t = 4$ days. Positive and negative returns are represented by the blue and orange lines respectively.

Asset, Time interval	Positive returns		Negative returns	
	Sample size	Power-law fitted exponent	Sample size	Power-law fitted exponent
S&P 500, $\Delta t = 1$ min	1,526,954	3.78 ± 0.03	1,232,069	3.92 ± 0.05
S&P 500, $\Delta t = 1$ d	1,252	3.87 ± 0.2	1,011	3.18 ± 0.2
S&P 500, $\Delta t = 4$ d	344	3.6 ± 0.3	221	2.7 ± 0.2
S&P 500, $\Delta t = 10$ d	150	4.0 ± 0.5	76	3.5 ± 0.7
NIKKEI, $\Delta t = 1$ min	2,246,546	4.06 ± 0.02	1,517,898	4.02 ± 0.03
NIKKEI, $\Delta t = 1$ d	1,722	4.1 ± 0.9	1,945	4.7 ± 0.3
NIKKEI, $\Delta t = 4$ d	411	4.1 ± 0.5	505	3.7 ± 0.3
NIKKEI, $\Delta t = 10$ d	152	3.62 ± 0.5	214	4.4 ± 0.6
Bitcoin, $\Delta t = 1$ min	204,336	3.64 ± 0.03	188,917	3.84 ± 0.04
Bitcoin, $\Delta t = 1$ d	140	3.0 ± 0.6	134	4.4 ± 0.7
Bitcoin, $\Delta t = 4$ d	38	3.0 ± 0.6	31	4 ± 1

TABLE 1. Sample sizes of the positive and negative return distributions and power-law fitted exponents of the positive and negative tails. The standard uncertainty of the exponents was determined by taking the standard deviation of alpha during power-law fitting.

Considering that Gaussian distribution is a special case of Levy stable distributions where $\alpha = 2$, the exponents computed in Table 1 suggest that the return distributions across the various time scales Δt chosen are distinctly non-Gaussian. As the α across time scales are greater than 2, returns exhibit unstable distributions, which agrees with Gopikrishnan et al (1999). The functional forms of the distributions will hence change with the number of samples. The small range of exponent values across time scales Δt within each type of financial data (3.78 ± 0.03 to 4.0 ± 0.5 for S&P, 3.62 ± 0.5 to 4.7 ± 0.3 for NIKKEI, and 3.0 ± 0.6 to 4.4 ± 0.7 for Bitcoin) attests to the time-scaling of returns observed by Mandelbrot (1963) [6] and Gopikrishnan et al (1999) [3].

Figure 3 shows the complementary cumulative probability distribution of Bitcoin returns at $\Delta t = 1s$. The clear linear relationship on the log-log plots of the returns distribution that was observed in Figure 2 is absent at this frequency. A possible reason is that the returns distribution in such small time-resolutions could be dominantly influenced by noise from discrete tick transactions and internet or platform latency whereas when data frequency $\Delta t \leq 1$ min, the noise gets averaged out and the returns form a smooth power-law tail. Since the return distribution of Bitcoin at $\Delta t = 1s$ does not exhibit asymptotic power-law behaviour, we did not pursue further analysis of the moments and the Levy-fitted parameters, whose hypotheses were conditional on the presence of power-law tails.

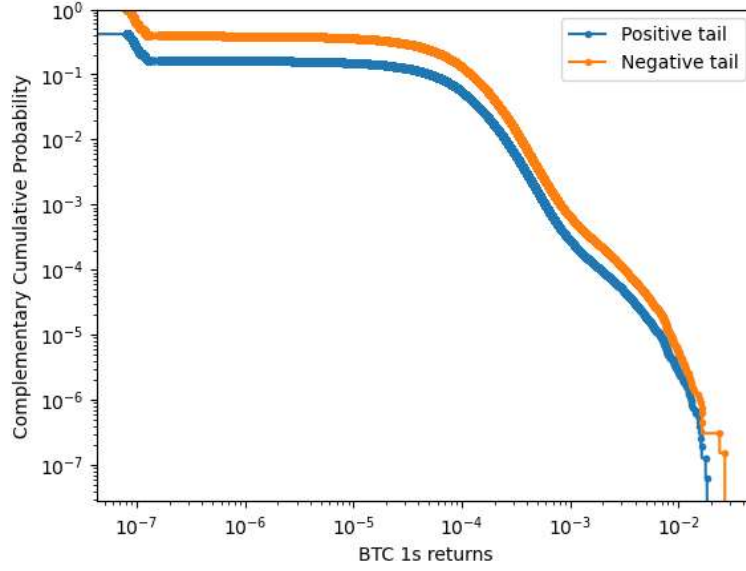
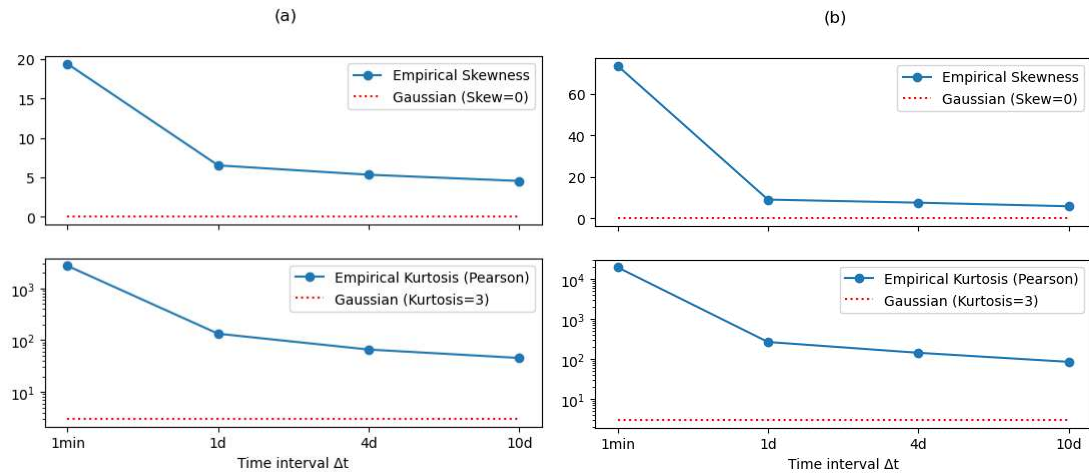


FIGURE 3. Log-log plots of the complementary cumulative probability distributions of Bitcoin returns when $\Delta t = 1s$. Positive and negative returns are represented by the blue and orange lines respectively.

The third and fourth moments of the S&P 500, NIKKEI, and Bitcoin data across time scales are plotted in Figure 4. Note that although the third moment of Bitcoin returns appears to have already converged with Gaussian distribution at $\Delta t = 4d$, the fourth moment has not, hence the Bitcoin returns distribution overall has yet to attain convergence to Gaussian at that time interval. Studying the three types of returns, we observe a gradual convergence to a Gaussian distribution with from $\Delta t = 4d$ onwards, in agreement with Gopikrishnan et al's (1999) findings.



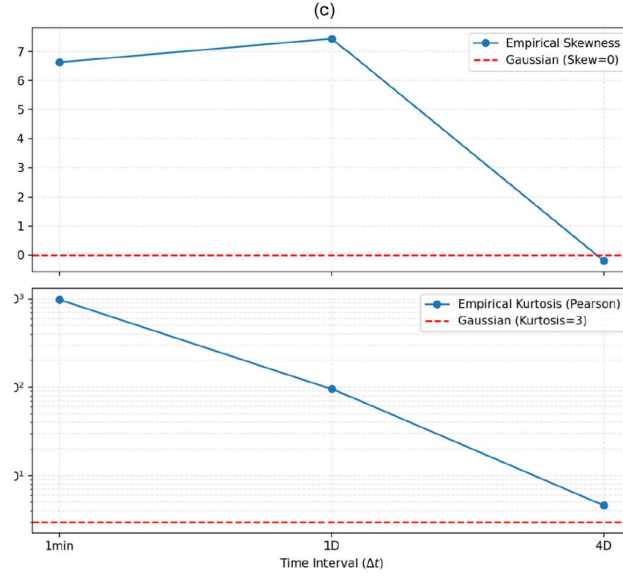


FIGURE 4. Skewness or third moment (top) and kurtosis or fourth moment (bottom) plots of (a) S&P 500, (b) NIKKEI, and (c) Bitcoin at different time intervals Δt . The red dashed line indicates the third and fourth moments of a Gaussian distribution, which are equal to 0 and 3 respectively.

The estimated stability α and asymmetry β parameters for various days up to 60 days prior to the crisis are plotted in Table 2. Crises were classified into critical events and crashes in the same way as Bielinskyi et al (2019). While critical events are significant falling of price, crashes are characterized by much larger losses concentrated over a short time. The absolute values of the α and β parameters for S&P 500 and NIKKEI are close to those computed by Bielinskyi et al. Interestingly, for Bitcoin, the absolute values of β are noticeably smaller than S&P 500 and NIKKEI and instead similar to the stock markets of developing economies, but the values of α are roughly the same as S&P and NIKKEI and markedly higher than those of developing economies. This suggests that large movements in Bitcoin price occur in both directions at approximately equal probability and that Bitcoin prices possess a thinner tail and lower probability of extreme returns. As the date of crisis approaches, the absolute values of the α and β parameters for both Bitcoin crashes exhibit a generally decreasing trend, agreeing with the observations made by Bielinskyi et al (2019) on DJIA. However, the parameters for S&P 500 and NIKKEI did not yield this decline in the 60 days before their respective crises. This shows that the ability of Levy stable distribution parameters in indicating the arrival of crises is not robust in all markets and events.

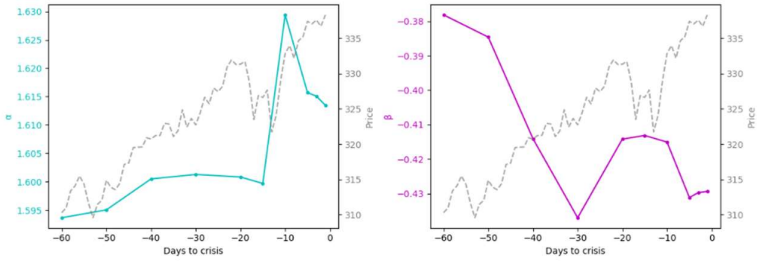
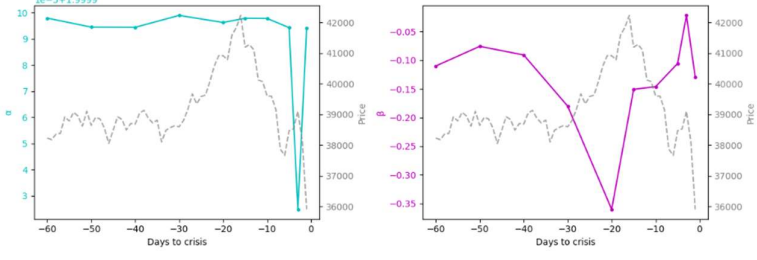
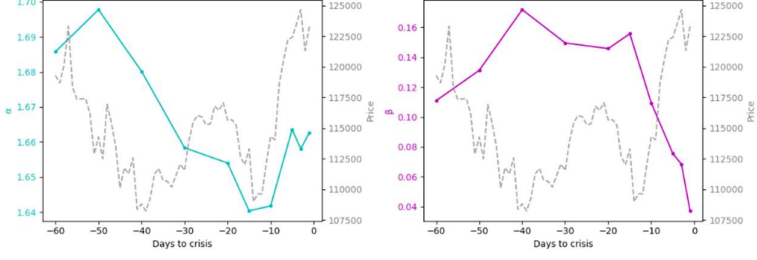
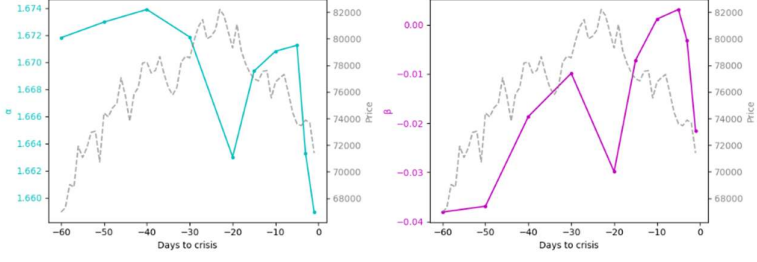
Financial data	Date of crisis	Estimated parameters of Levy stable distribution	
S&P 500	20 February 2020 (Crash)		
NIKKEI	5 August 2024 (Critical event)		
Bitcoin	10 October 2025 (Crash)		
Bitcoin	3 June 2026 (Crash)		

TABLE 2. First column states the type of financial data being analyzed. Second column states the date of the financial crisis affecting each type of financial data. Third column contains plots of the stability α (cyan), asymmetry β parameters (magenta), and prices (grey) up to 60 days prior to the crisis.

6. Conclusion

Our empirical analyses were in concurrence with Gopikrishnan et al (1999) that the asymptotic distributions of returns for groups of assets (represented by S&P 500 and NIKKEI) and for individual assets (Bitcoin) across time scales are consistent with a power-law, at $\Delta t = 1$ min

up to 10 days for S&P and NIKKEI and $\Delta t = 1$ min up to 4 days for Bitcoin. Fitting the empirical data to a power-law fit yields exponents $\alpha > 2$ for time intervals $\Delta t = 1$ min and longer, thus the tails of the distribution of returns cannot be entirely modelled on Levy stable distribution. A gradual convergence of the distribution of returns towards a Gaussian form was observed when increasing the time scale Δt , exhibiting a good fit to the truncated Levy distribution. By approximating the returns as a Levy stable distribution and estimating the stability α and asymmetry β parameters over rolling 500-day windows, we found evidence that the parameters can be used as precursor-indicators to market crises for Bitcoin. However, this may not be applicable to other markets or different circumstances, such as with the S&P 500 and NIKKEI crises. A broader investigation into the pre-crisis behaviour of α and β for various stock and cryptocurrency markets is needed to explore the extent to which these parameters can be relied on as warning signals of a market crash.

Overall, these findings highlight the importance of power-law to Econophysics in understanding the complex dynamics of financial markets. By implementing insights from statistical physics and complexity theory, Econophysics provides a framework that enables characterization of the probability distribution of financial returns and potential development of predictive tools able to identify the onset of financial crises.

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